Quadratics in factored form

1 For each of the following equations:

- **i** Find the y value of the y-intercept.
- **ii** Find the x values of the x-intercepts.
- **iii** State the equation of the axis of symmetry.
- **iv** Find the coordinates of the vertex.
- **v** Plot the graph of the parabola.

a $y = x(x-4)$

b $y = x(6-x)$.

c $y = (x-5)(x-3)$

d $y = (x+4)(x-2)$

e $y = (1-x)(x-5)$

f $y = (5-x)(3-x)$.

2 Consider the given equations:

- **i** Determine the y value of the y-intercept of the graph.
- **ii** Determine the x values of the x-intercepts.
- **iii** Complete the table of values for the equation.
- **iv** Determine the coordinates of the vertex of the parabola.
- **v** Graph the parabola.

a $y = x(x+6)$

x	-5	-4	-3	-2	-1
y					

b $y = (x-2)(x-6)$.

x	0	2	4	6	8
y					

c $y = (2-x)(x+4)$

x	-5	-3	-1	1	3
y					

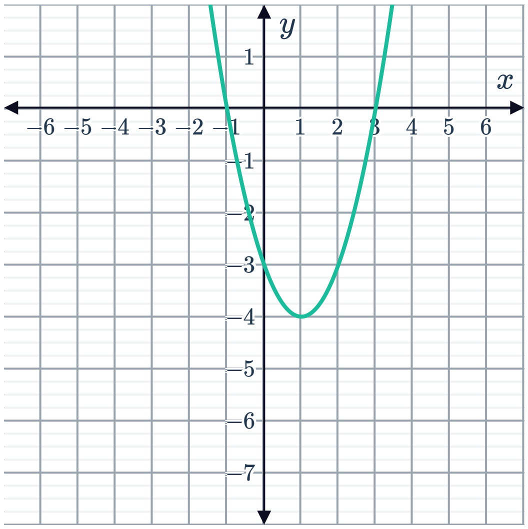
3 Consider the function $y = (x+5)(x+1)$.

a Plot the x-intercepts and y-intercept of the function, and use these points to plot the graph of $y = (x+5)(x+1)$ on the number plane.

b Determine the y value of the y-intercept of $y = -(x+5)(x+1)$.

4 Consider the following graphs:

A



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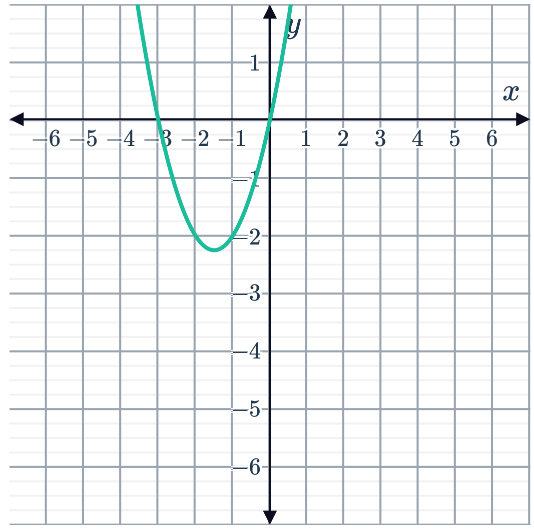

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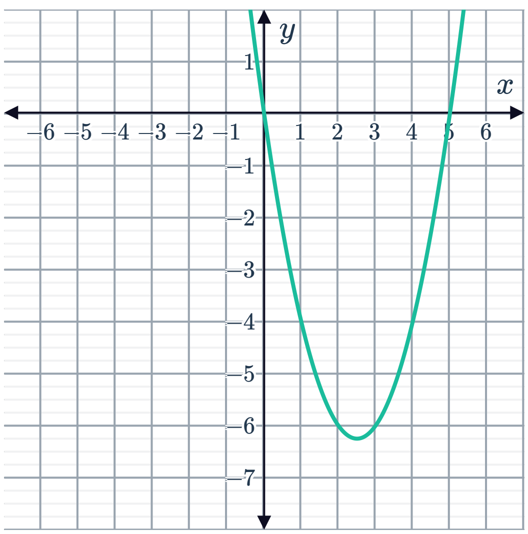

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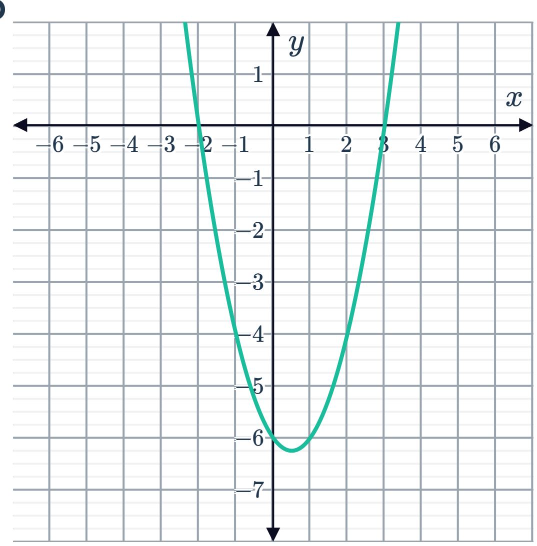


Match each of the following equations to the correct graph:

a $y = x\left(x+3\right)$

b $y = x\left(x-5\right)$

c $y = \left(x+2\right)\left(x-3\right)$

d $y = \left(x-3\right)\left(x+1\right)$

5 Consider the equation of a parabola of the form $y = \left(x-a\right)\left(x-b\right)$.

a The parabola has x-intercepts at $x=-1$ and $x=-5$. Write its equation.

b Determine the y value of the y-intercept of the parabola.

c Use the intercepts to plot the graph of the parabola.

6 Consider a quadratic function of the form $y = (x - a)(x - b)$.

a The function has x-intercepts at $x = \sqrt{3}$ and $x = -\sqrt{3}$. Write its equation in standard form.

b What is the y value of function when $x = 1$?

c Plot the graph of the function.

7 Consider the following equations:

i Factor the quadratic.

ii Find the x values of the x-intercepts.

iii Find the x value of the vertex.

iv Find the y value of the vertex.

v Plot the graph of the function.

a $y = 6x - x^2$ b $y = x^2 + 2x - 8$

8 Consider the equation $y = 4x + x^2$.

a Factor the expression $4x + x^2$.

b Find the x values of the x-intercepts.

c Find the y value of the y-intercept.

d Complete the table of values for the equation.

x	-4	-3	-2	-1	0
y					

e Determine the coordinates of the vertex.

f Graph the function.

9 Consider the equation $y = x^2 + 8x + 12$.

a Factor the expression $x^2 + 8x + 12$.

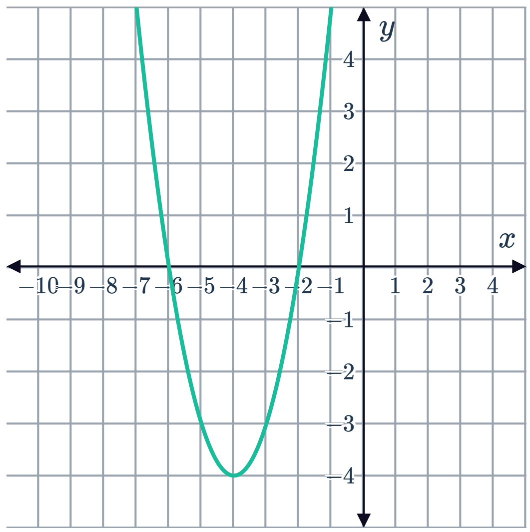
b Solve the x values of the x-intercepts of the quadratic function $y = x^2 + 8x + 12$.

c Complete the table of values for the equ.

x	-8	-6	-4	-2	0
y					

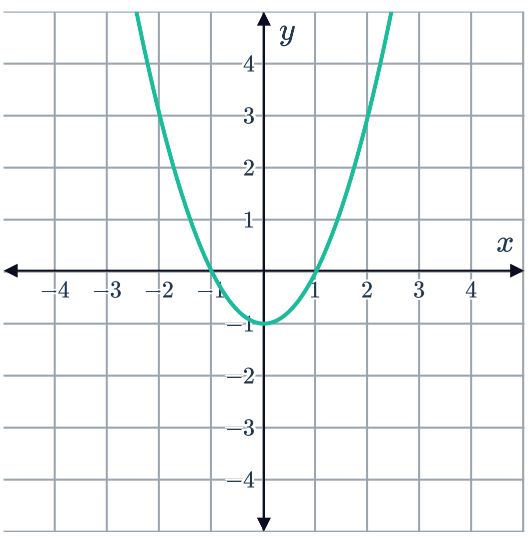
d Determine the coordinates of the vertex.

e Graph the function.



10 State whether the following graphs represent a parabola of the form $y = (x - a)^2$?

a



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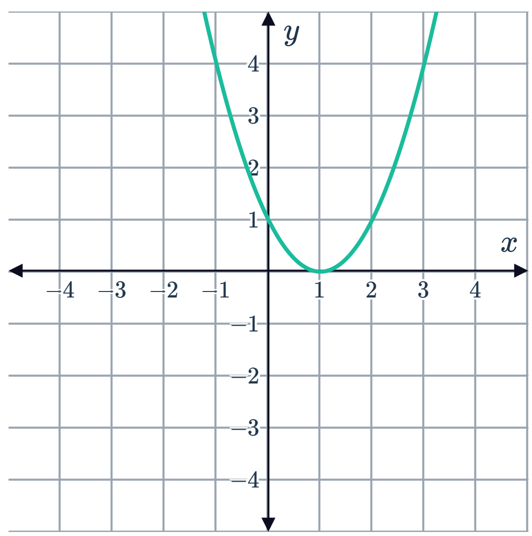
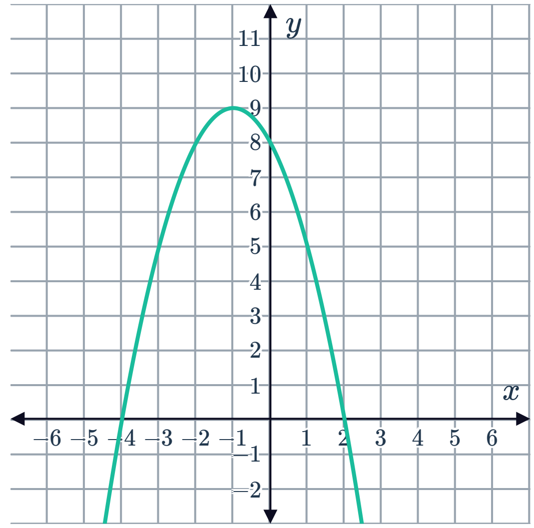


11 Consider the graph.



a State the y value of the y-intercept of the graph.

b Write the equation of this graph in factored form.

12 State the the axis of symmetry of the parabola $y = k\left(x - 7\right)\left(x + 7\right)$ for any value of k.

13 A parabola has its vertex at $x = -5$ and one of the x-intercepts is $x = 1$.

a Find the x value of the other x-intercept?

b If it has a y-intercept at -11, determine the equation of the parabola.

c Find the coordinates of the vertex.

14 Consider the equation $y = \left(x - b\right)\left(x + b\right)$.

a Write the equation in standard form.

b Find the value of a such that the parabola $y = x^2 + ax - 9$ will have x-intercepts that are equidistant from the origin.

15 A football is kicked into the air and its height h meters above the ground at time t seconds after being kicked is given by $h = -t^2 + 20t$.

a Assuming the ball starts at height 0, determine the time t the ball will hit the ground.

b Find the greatest height the ball reaches above the ground.

Additional questions

16 State the factored form for a quadratic equation and describe its features.

17 Describe the axis of symmetry of a quadratic function.

18 For each of the following quadratic equations, state the x-values of the x-intercepts:

a $y=(x-1)(x-4)$

b $y=(x-3)(x+7)$

c $y=x(x-4)$

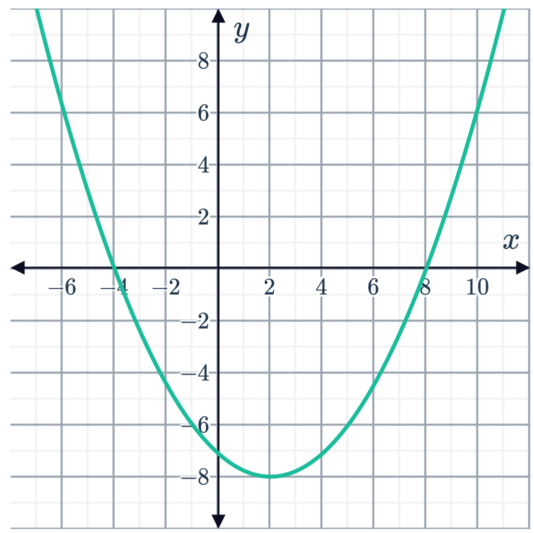
d $y=2(x-1)(x+5)$

19 For each of the following quadratic functions, calculate the y-value of the y-intercept:

a $y=3(x-1)(x-8)$

b $y=x(x+9)$

20 Identify the axis of symmetry of this quadratic function.



21 For each quadratic function:

i State the coordinates of the x-intercepts.

ii Determine the axis of symmetry.

iii Determine the coordinates of the vertex.

iv Draw the graph of the function.

a $y=(x-2)(x+4)$

b $y=-2(x-1)(x-5)$

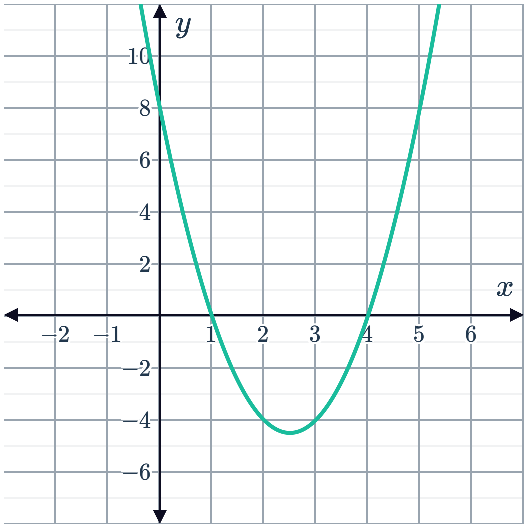
22 For each graph:

i Identify the x-values of the x-intercepts.

ii Identify the y-value of the y-intercept.

iii Write the equation of the quadratic function in factored form.

a



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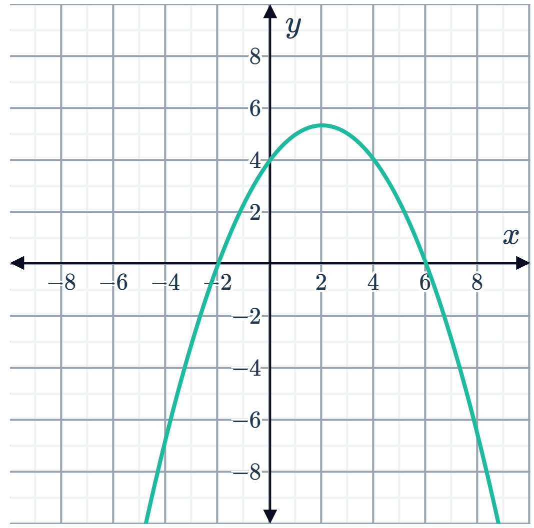
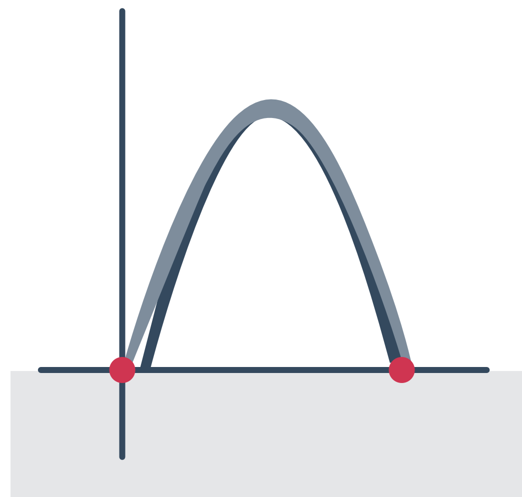


23 A quadratic function has x-intercepts of $(5, 0)$ and $(-3, 0)$, and a y-intercept of $(0, 60)$.
Write the equation of the function in factored form.

24 A quadratic function passes through the points $(-2, 0)$, $(9, 0)$ and $(0, -6)$.
Write the equation of the function in factored form.

25 The x-values of the x-intercepts of a quadratic function are $x = -4$ and $x = -7$. The graph of the function passes through the point $(-3, 12)$.
Write the equation of the function in factored form.

26 Xia notices that the Sunshine State Arch is in the shape of a quadratic function. They know that the arch has a height of 110 ft and the feet of the arch are 100 ft apart. Xia chooses to let the x-intercepts of the arch be the origin and $(100, 0)$.



a Determine the coordinates of the vertex of the arch.

b Write the equation of the quadratic function for Xia's representation of the Sunshine State Arch in factored form.

c Find the domain and range of the Sunshine State Arch.

27 Ami throws a javelin forwards in a parabolic arc from the ground. Using photos that her friend Maryellen is taking from the stands, she determines that 10 yards horizontally from where she threw the javelin, it reaches a maximum height 5 yards above the ground. Ami models her throw with the origin of a cartesian plane being the point 20 yards behind her.

a For Ami's model, state the coordinates of the x-intercepts.

b Write the equation for the quadratic function modelling the path of Ami's throw in factored form.

28 Wilson tosses an eraser into the air and counts how long it takes for the eraser to return to his hand. He estimates that it takes 2 seconds and that he is tossing the eraser 6 ft into the air.
Wilson models the height of the eraser above his hand in feet as a function h of time t in secondes, letting $t = 0$ be when he tosses the eraser and $h = 0$ be the height of his hand.

a Explain how to find the intercepts of the function and state them.

b Assuming that the path of the eraser is symmetric going up and coming down, state the coordinates of the vertex of the function.

c Draw a graph of the function.

d Write the equation of the function in factored form.

29 A quadratic function has the factored form e[redacted]on $y = k(x - x_0)(x - 12)$.
If the function has a vertex at the point $(8, 15)$, determine the values of x_0 and k. Justify your answers.